Probabilistic Methods

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§1 Problems

§1.1 Level 1

1. A biased coin has probability of head $= p$ and tail $= 1 - p$. The coin is tossed n times successively. Find the expected number of heads.

2. Two hundred students participate in a mathematical contest. They had six problems to solve. It is known that each problem was correctly solved by at least 120 participants. Prove that there must be two participants such that every problem was solved by at least one of these two students.

3. In the Duma there are 1600 delegates, who have formed 16000 committees of 80 persons each. Prove that one can find to committees having no fewer than four common members.

§1.2 Level 2

1. In a class of n students, each boy knows at least one girl. Show that one can choose a group of more than half of the students such that every boy in the group knows an odd number of girls in the group.

2. Consider n real numbers, not all zero, with sum zero. Prove that one label the numbers as $a_1, a_2, \ldots, a_n$ in some order such that

$$a_1 a_2 + a_2 a_3 + \cdots + a_n a_1 < 0.$$

3. [***Szele's Theorem***] Show that for every n, there exists a tournament on n vertices having at least $n!/2^{n-1}$ Hamiltonian paths.

4. Let A be a set of N residues modulo N^2. Prove that there exists a set B of N residues modulo N^2 such that the set $S = A + B$ contains at least half of the residues modulo N^2. Here,

$$A + B = \{a + b \mid a \in A, b \in B\}$$

5. Say all the set of natural numbers is partitioned into n disjoint APs with common differences $r_1, r_2, \ldots, r_n$. Show that
$$\frac{1}{r_1} + \frac{1}{r_2} + \cdots + \frac{1}{r_n} = 1.$$

§1.3 Level 3 (based on Finite sets)

1. [***Lubell-Yamamoto-Meshalkin Inequality***] Let $A_1, A_2, \ldots, A_s$ be subsets of $\{1, 2, \ldots, n\}$ such that A_i is not a subset of A_j for any i and j. Let a_i denote $|A_i|$ for each i. Show that
$$\sum_{i=1}^{s} \frac{1}{\binom{n}{a_i}} \leq 1.$$

2. [***Sperner's Theorem***] Let $A_1, A_2, \ldots, A_s$ be subsets of $\{1, 2, \ldots, n\}$ such that A_i is not a subset of A_j for any i and j. Such a family of sets is known as an *antichain*. Show that
$$s \leq \binom{n}{\lfloor n/2 \rfloor}.$$

3. [***Bollobas's Theorem***] Let $A_1, A_2, \ldots, A_n$ be sets of cardinality s, and let $B_1, B_2, \ldots, B_n$ be sets of cardinality t. Further suppose that $A_i \cap B_j = \emptyset$ if and only if $i = j$. Show that
$$n \leq \binom{t+s}{s}.$$

§1.4 Level 4

1. Let S be a finite set of points in the plane such that no three of them are collinear. For each convex polygon P whose vertices are in S, let $a(P)$ be the number of vertices of P, and let $b(P)$ be the number of points in S which are outside P. A line segment, a point, and the empty set are considered as convex polygons of 2, 1, and 0 vertices respectively. Prove that for every real number x,
$$\sum_{P} x^{a(P)}(1-x)^{b(P)} = 1,$$
where the sum is taken over all convex polygons with vertices in S.